Muhammad Junaid Iqbal

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Computer Engineering student with hands-on experience in AI/ML, computer vision, embedded systems, and database-driven applications. Project work spans deep learning, retrieval-augmented generation (RAG), signal processing, and full-stack GUI/database development. Seeking to apply and grow my skills in AI, embedded systems, and software development.

EDUCATION

Bachelor of Computer Engineering, Capital University of Science and Technology (CUST), Islamabad — 2023 – Present (Expected 2027)

Coursework: Artificial Intelligence, Data Structures and Algorithms, DBMS, Operating Systems, Computer Networks, OOP, Software Engineering

EXPERIENCE

Machine Learning Intern, Ezitech (Remote) — 13 Jul 2026 – 21 Aug 2026 | Certificate ID: ETI-INT-08-26-28501

- Built the Enterprise Knowledge Intelligence Engine (EKIE), a production-grade RAG platform with a two-person intern team over a four-week case study.
- Implemented cross-encoder reranking, document comparison and summarization, and Neo4j knowledge graph relationship inference with confidence scoring and explainability.
- Reached recall@10 = 1.0 and MRR = 0.93 on a 40-question eval set; 333 passing tests; documented 39 API routes with Swagger; containerized with Docker.

ACADEMIC PROJECTS

AI-Driven Bidirectional Sign Language Communication System (In progress) — Python, Deep Learning, Computer Vision

- Building a real-time, two-way system combining gesture recognition with speech and text conversion using deep learning and computer vision.

Stock Price Prediction Using LSTM Networks — Python, TensorFlow, Keras

- Built an LSTM model to forecast stock price trends from preprocessed, normalized time-series data.

Supermarket Management System — Python, Oracle SQL, Tkinter

- Built a GUI inventory and customer management app on a relational database with CRUD, role-based login, and automated reports.

Smart Parking Slot Management System — STM32, Embedded C

- Built a smart parking slot prototype on an STM32 microcontroller with real-time keypad input handling.

LMS-Based Noise Cancellation for Industrial Vibration — MATLAB, TMS320C6713 DSK

- Implemented an LMS adaptive filter for industrial vibration noise cancellation; benchmarked against FIR and IIR filters in MATLAB and on a DSP kit.

TECHNICAL SKILLS

- **Languages:** Java, Python, C, Embedded C, SQL, MATLAB, JavaScript
- **AI/ML & Backend:** TensorFlow, Keras, LSTM, RAG, Cross-Encoder Reranking, Gemini API, ChromaDB, Neo4j, Swagger, pytest
- **Embedded & Databases:** STM32, TMS320C6713 DSK, MySQL, Oracle SQL Developer
- **Tools:** Docker, Git, Jupyter Notebook, Keil uVision, Code Composer Studio, STM32CubeIDE